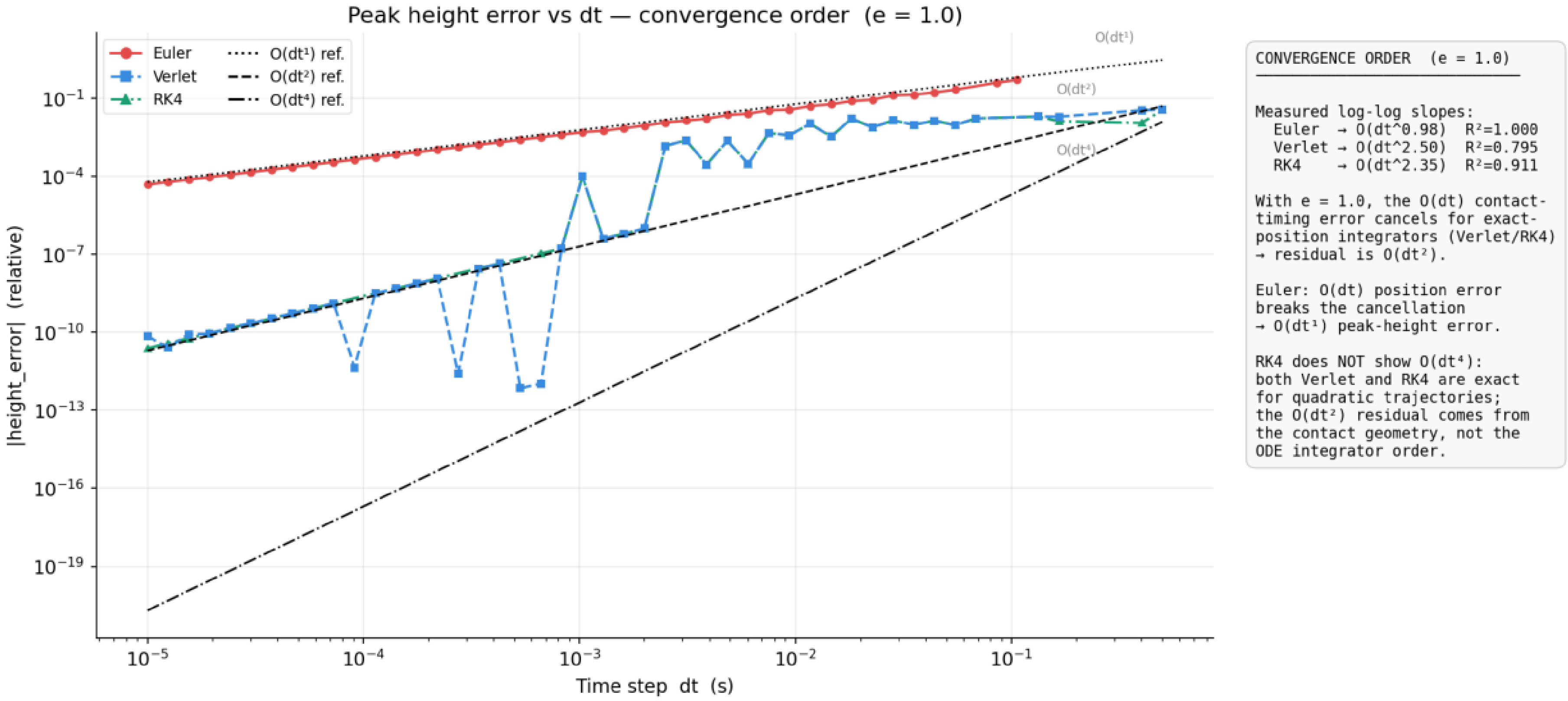
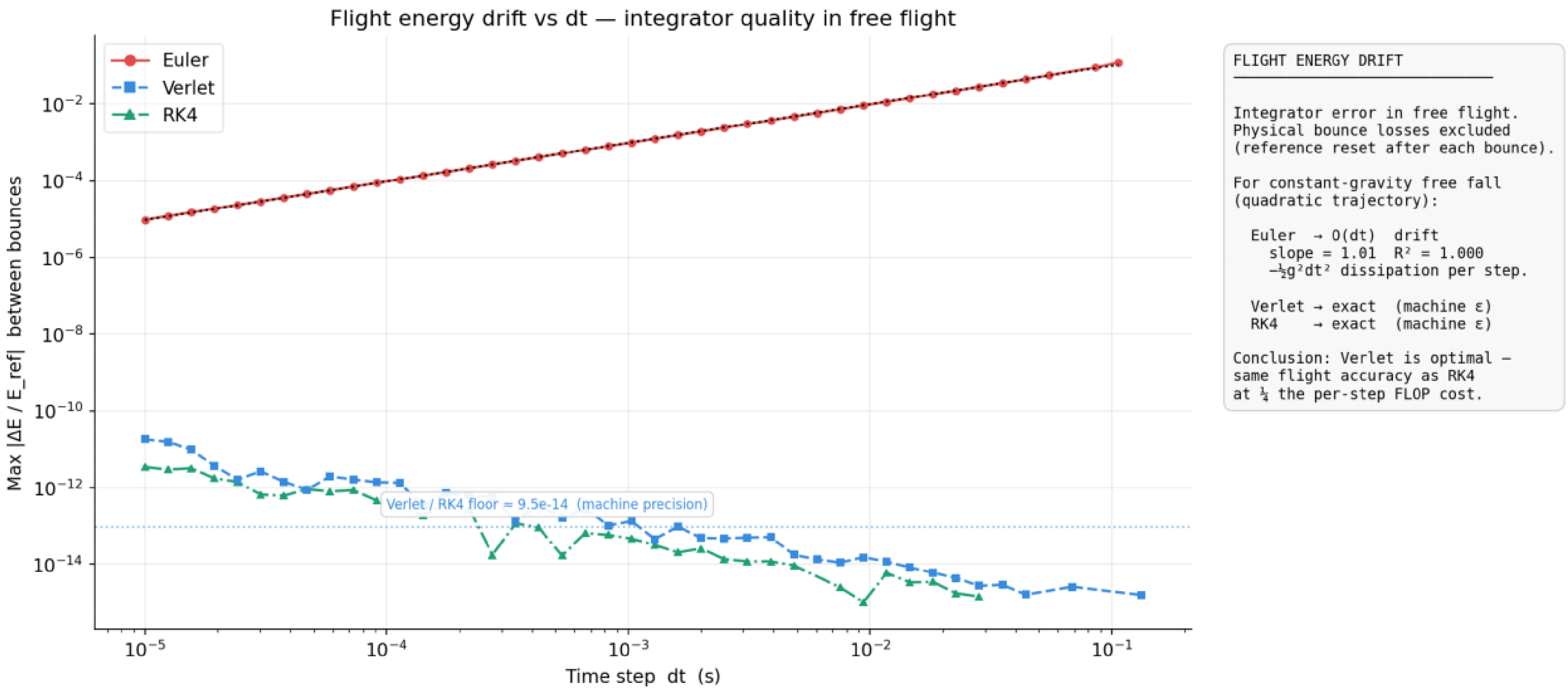
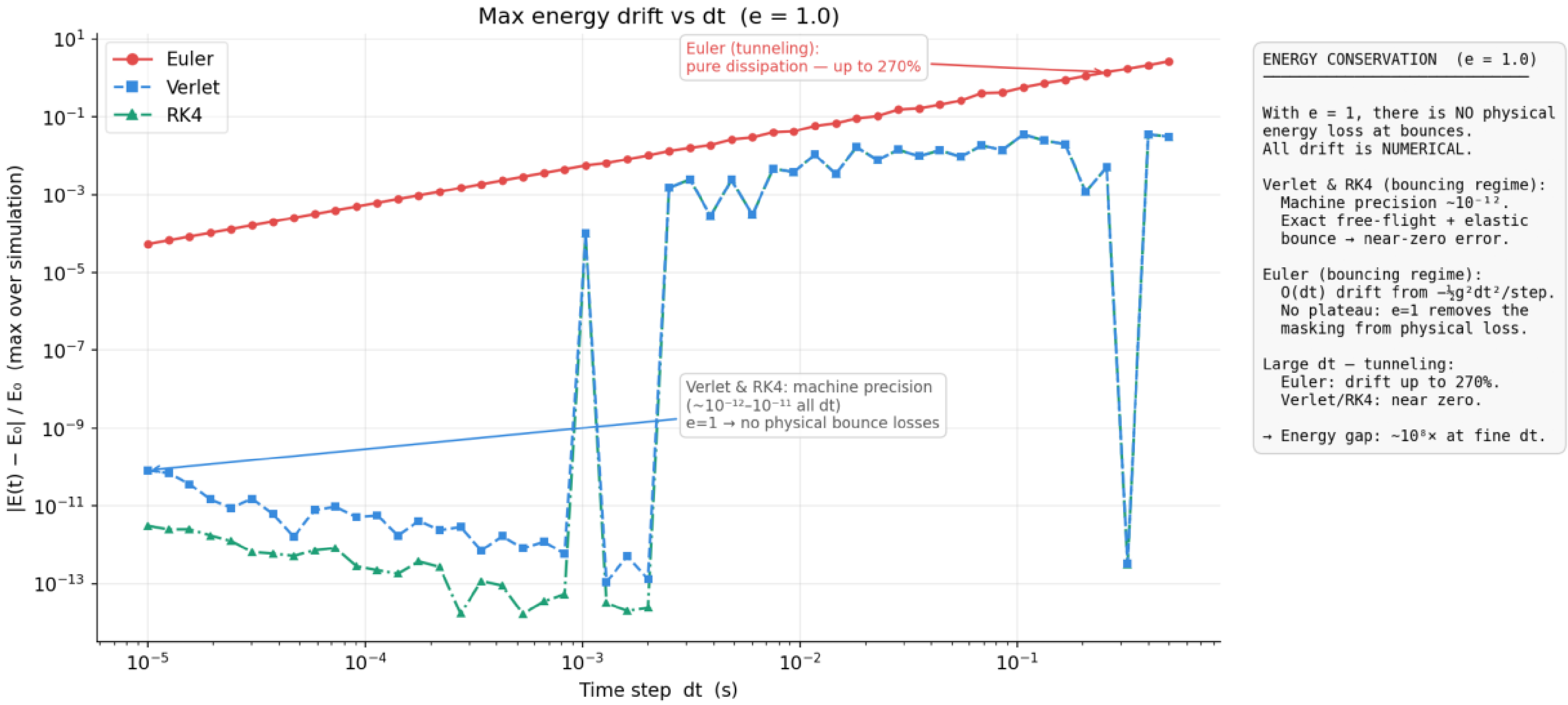
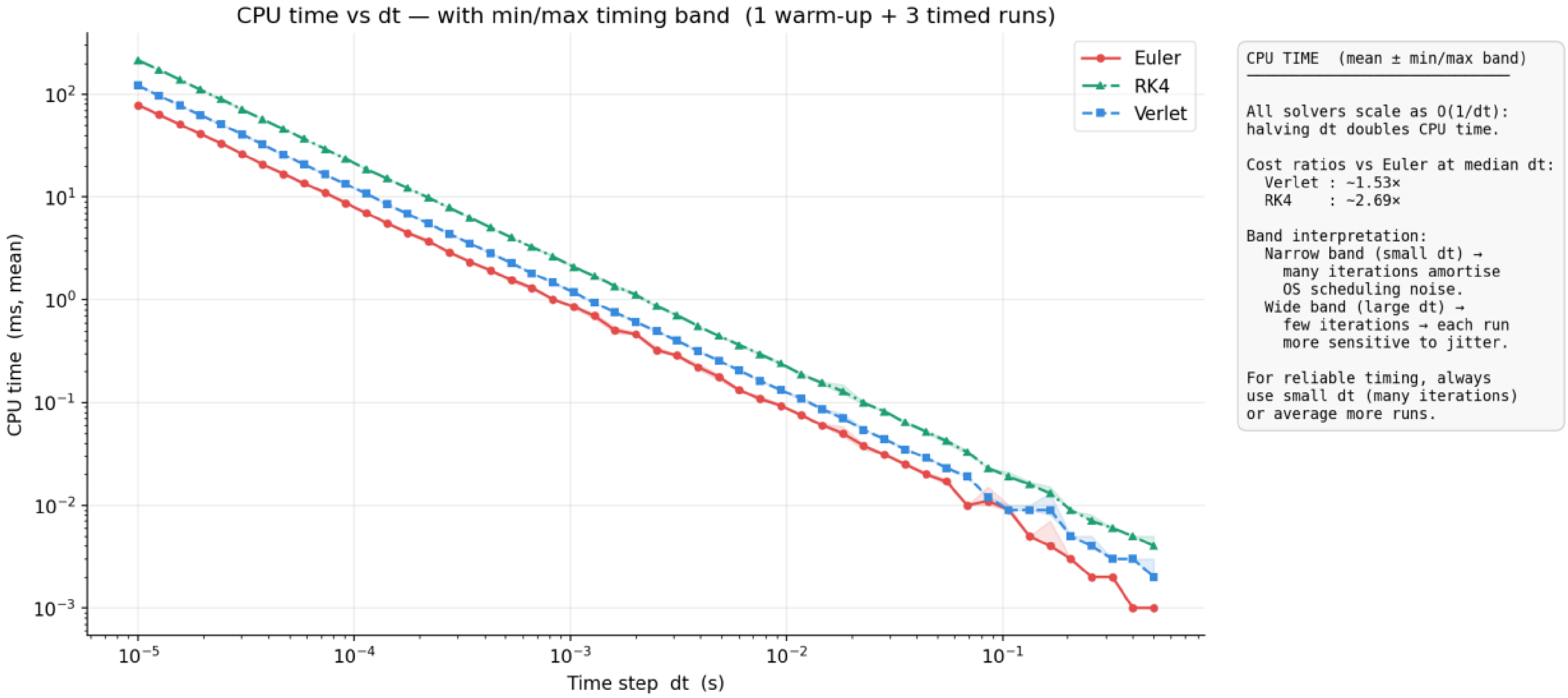

Peak height error vs dt — convergence order (e = 1.0)
Euler
Verlet
RK4
O(dt¹) ref.
O(dt²) ref.
O(dt⁴) ref.
O(dt¹)
O(dt²)
O(dt⁴)
|height_error| (relative)
Time step dt (s)
CONVERGENCE ORDER (e = 1.0)
Measured log-log slopes:
Euler → O(dt^0.98) R²=1.000
Verlet → O(dt^2.50) R²=0.795
RK4 → O(dt^2.35) R²=0.911
With e = 1.0, the O(dt) contact-timing error cancels for exact-position integrators (Verlet/RK4) → residual is O(dt²).
Euler: O(dt) position error breaks the cancellation → O(dt¹) peak-height error.
RK4 does NOT show O(dt⁴): both Verlet and RK4 are exact for quadratic trajectories; the O(dt²) residual comes from the contact geometry, not the ODE integrator order.

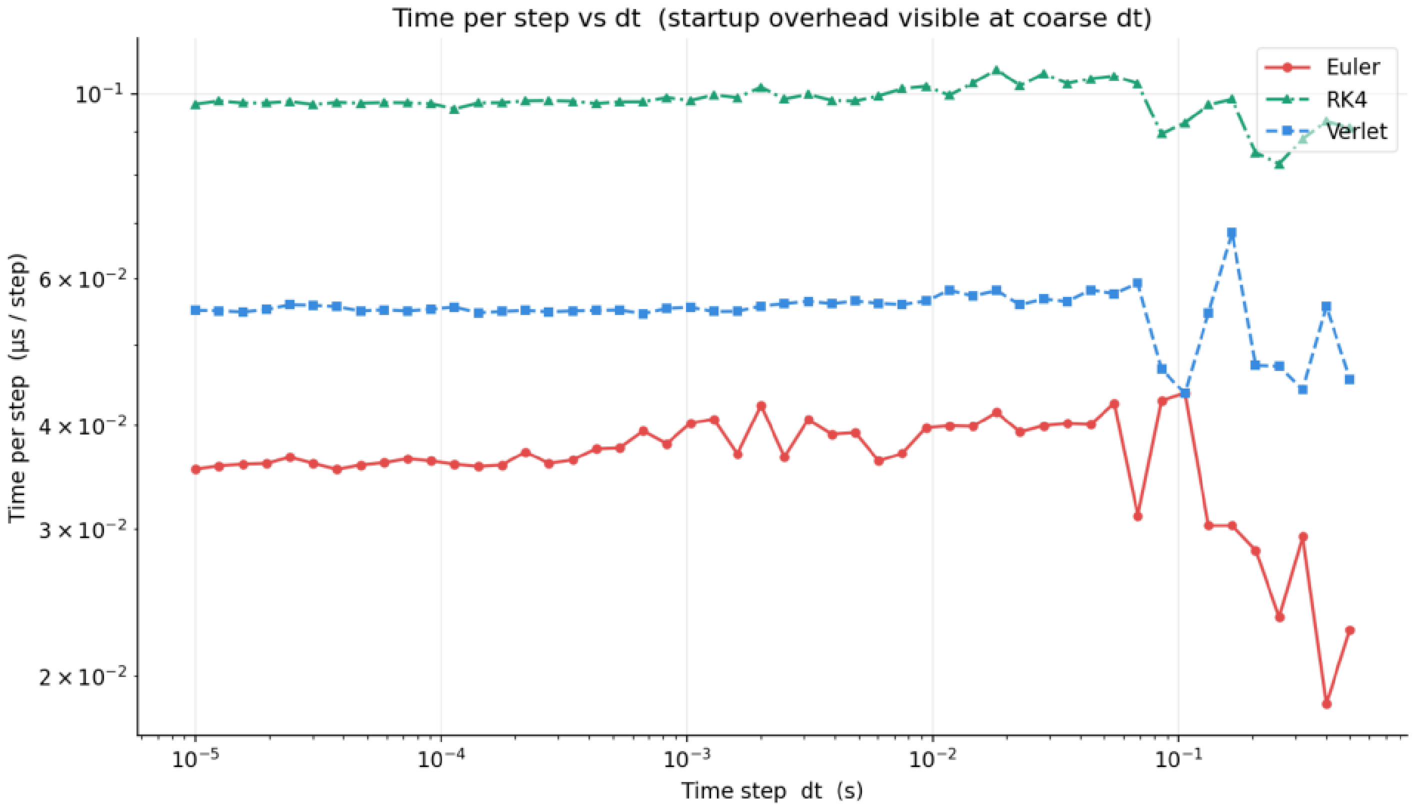

Flight energy drift vs dt — integrator quality in free flight
Euler
Verlet
RK4
Max |ΔE / E_ref| between bounces
Time step dt (s)
Verlet / RK4 floor ≈ 9.5e-14 (machine precision)
FLIGHT ENERGY DRIFT
Integrator error in free flight.
Physical bounce losses excluded
(reference reset after each bounce).
For constant-gravity free fall
(quadratic trajectory):
Euler → O(dt) drift
slope = 1.01 R^2 = 1.000
$-\frac{1}{2}g^2dt^2$ dissipation per step.
Verlet → exact (machine ε)
RK4 → exact (machine ε)
Conclusion: Verlet is optimal —
same flight accuracy as RK4
at ¼ the per-step FLOP cost.



Max energy drift vs dt (e = 1.0)
Euler
Verlet
RK4
Euler (tunneling):
pure dissipation — up to 270%
Verlet & RK4: machine precision
(~10^{-12}–10^{-11} all dt)
e=1 → no physical bounce losses
$|E(t) - E_0| / E_0$ (max over simulation)
Time step dt (s)
ENERGY CONSERVATION (e = 1.0)
With e = 1, there is NO physical
energy loss at bounces.
All drift is NUMERICAL.
Verlet & RK4 (bouncing regime):
Machine precision ~10^{-12}.
Exact free-flight + elastic
bounce → near-zero error.
Euler (bouncing regime):
O(dt) drift from −½g²dt²/step.
No plateau: e=1 removes the
masking from physical loss.
Large dt — tunneling:
Euler: drift up to 270%.
Verlet/RK4: near zero.
→ Energy gap: ~10^8× at fine dt.



CPU time vs dt — with min/max timing band (1 warm-up + 3 timed runs)
CPU time (ms, mean)
Time step dt (s)
Euler
RK4
Verlet
CPU TIME (mean ± min/max band)
All solvers scale as O(1/dt):
halving dt doubles CPU time.
Cost ratios vs Euler at median dt:
Verlet : ~1.53×
RK4 : ~2.69×
Band interpretation:
Narrow band (small dt) →
many iterations amortise
OS scheduling noise.
Wide band (large dt) →
few iterations → each run
more sensitive to jitter.
For reliable timing, always
use small dt (many iterations)
or average more runs.

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TIME PER STEP
─────────────────────────────

time_per_step = cpu_mean / N_steps

Plateau at fine dt = pure
integration cost (overhead
fully amortised over many steps).

Sustained per-step cost (median
of 10 finest-dt runs):
  Euler  : 0.036 μs/step
  Verlet : 0.055 μs/step
  RK4    : 0.098 μs/step

Rising curve at coarse dt:
  startup + teardown overhead
  dominates when only O(10)
  steps are executed.

Use fine-dt values for real-time
per-step budget planning.
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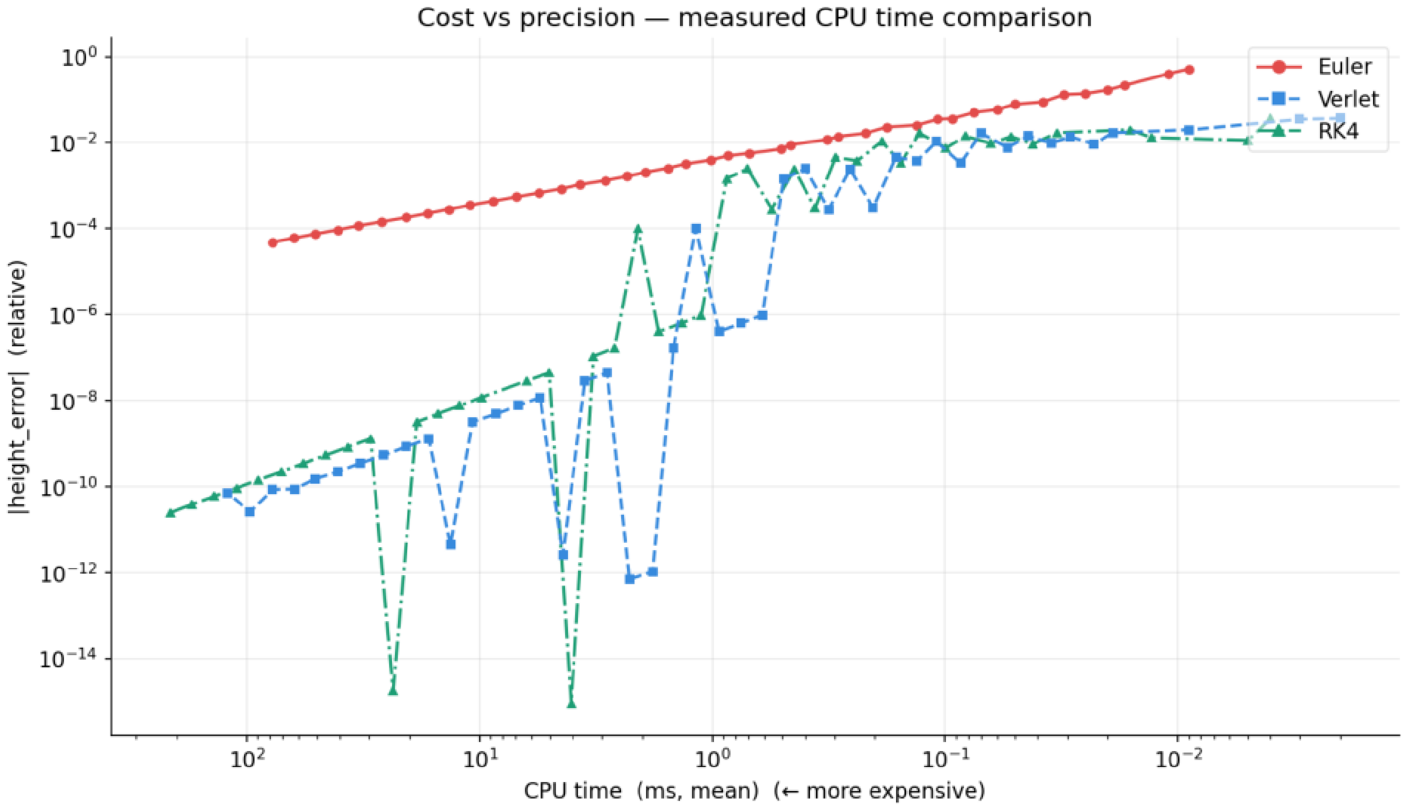


COST vs PRECISION

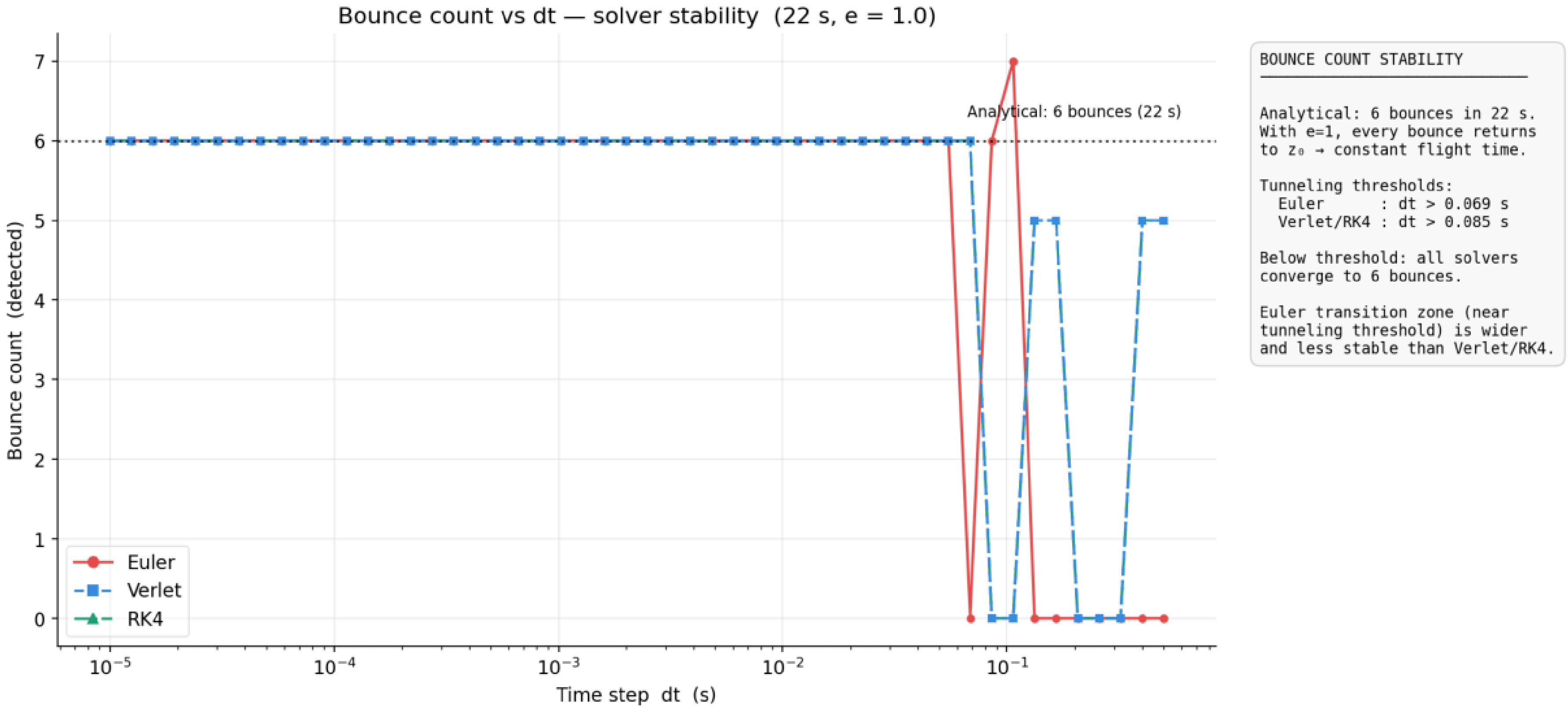
X = measured CPU time (ms).
Left = more expensive.

Euler:
O(dt¹) height error.

Verlet:
O(dt²) height error.
Same CPU cost as Euler
→ Strictly dominates Euler.

RK4:
O(dt²) height error.
~3.5× more expensive
than Verlet at same accuracy.

Verlet is uniquely optimal:
O(dt²) accuracy at O(dt¹)
cost — best achievable for
this contact model.


Bounce count vs dt — solver stability (22 s, e = 1.0)
Bounce count (detected)
Time step dt (s)
Analytical: 6 bounces (22 s)
Euler
Verlet
RK4
BOUNCE COUNT STABILITY
Analytical: 6 bounces in 22 s.
With e=1, every bounce returns
to z_0 → constant flight time.
Tunneling thresholds:
Euler : dt > 0.069 s
Verlet/RK4 : dt > 0.085 s
Below threshold: all solvers
converge to 6 bounces.
Euler transition zone (near
tunneling threshold) is wider
and less stable than Verlet/RK4.

E(t) time series — energy drift over simulation (e = 1.0)